MALARIA DETECTION USING CNN WITH GRAD-CAM

AI FOR ENGINEERS (UCS321)

Submitted By:

Vanshika (1024030163)
Vanshika (1024030166)
Nandini Garg (1024030169)
Aishwarya Verma (1024030170)
Diya Jain (1024031154)

Sub-Group:
2C16

Submitted To:
Dr. Jinee Goyal

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Thapar Institute of Engineering and Technology

ABSTRACT

The project is about the creation of an automated diagnostic tool using deep learning for the detection of malaria in red blood cell microscopy images. The system solves the limitations of standard manual microscopy such as image noise and class variability using a Convolutional Neural Network (CNN) trained with the National Institutes of Health (NIH) dataset of 27,558 images with an equal distribution of 13,779 parasitized and 13,779 uninfected cells. The preprocessing pipeline includes resizing images to 64 x 64 pixels, normalization and a lot of data augmentation (rotations, flips, zooms) to make the model robust. The model uses three convolutional blocks with batch normalization and max-pooling, followed by a dropout layer to prevent overfitting, achieving an accuracy of over 95%. The project involves Grad-CAM (Gradient-weighted Class Activation Mapping) to bridge the gap between AI performance and clinical trust. This explainability technique produces heatmaps that indicate the particular regions within a cell that led to the model's diagnosis, offering visual confirmation of the decision-making process for medical experts. The results show the combination of high accuracy CNNs and interpretable visual cues as a reliable and transparent solution for AI-assisted medical screening.

INTRODUCTION

Malaria continues to be a major global health threat, traditionally diagnosed by the labor intensive and error prone manual microscopic examination of blood smears. This project proposes an automated solution by training a Convolutional Neural Network (CNN) on the NIH dataset which contains 27,558 cell images. The pipeline uses careful preprocessing and data augmentation to be robust to noise and variations in images. The model obtained from this is over 95% accurate. This can be a quick and consistent screening tool to ease the work of laboratory professionals to a great extent and reduce the delay in diagnosis. An important aspect of this AI algorithm system is the use of Grad-CAM, which makes the AI algorithm a source of "explainable AI". Through heatmaps that indicate the areas in which the AI algorithm can see signs of infection, the AI algorithm system becomes more reliable as an explanation for why the predictions have been made. This allows the system to move from a "black box" to a trustworthy decision support system.

METHODOLOGY

The methodology for this project involves a structured pipeline ranging from data acquisition to model explainability. It is designed to ensure high classification accuracy while providing visual justifications for medical diagnostics.

1. Data Acquisition and Preparation :
The experiment uses the NIH Malaria Dataset provided by Kaggle, consisting of 27,558 cell images.

Data Balancing : The data is perfectly balanced with 13,779 images per each class of "Parasitized" and "Uninfected."

Preprocessing : All images were loaded with the help of OpenCV, resized to 64x64 pixels, and changed into RGB format. Then their pixel values were normalized between 0 and 1 to achieve fast convergence.

Data Augmentation : To generalize better, various techniques such as image rotation up to 20°, horizontal and vertical flipping, 15% zooming, and width/height shifting were performed.

2. CNN Architecture Design :
The main components of the model are the CNNs that form a sequential architecture and extract hierarchically structured features from the images of cells:

Convolutional Layers : Three main groups of 32, 64, and 128 kernels correspondingly. All the layers use the 3x3 kernel and apply the ReLU activation function.

Regularization and Pooling : Batch Normalization and 2x2 Max-Pooling operations are added to each convolution to improve its performance.

Classification : The extracted features are then flattened and fed into the Dense layer with 128 neurons and subsequently into the Dropout layer (0.5). Finally, the results are passed to the output Sigmoid layer.

3. Training and Evaluation :

Optimization : This is achieved through the use of the Adam Optimizer and Binary Crossentropy Loss Function.

Validation : Validation is done by setting aside a percentage of the data during the training process in order to test the performance of the model and to avoid memorization of the training dataset.

Evaluation Metrics : Evaluation of the performance of the model is carried out using the Confusion Matrix, Accuracy, Precision, Recall, and F1-Score measures.

4. Explainability with Grad-CAM :

The final stage involves the integration of Gradient-weighted Class Activation Mapping (Grad-CAM) :

Feature Mapping: It calculates the gradients of the last layer with respect to the output classes.

Creation of Heatmap: By taking the weighted average of the feature maps, it creates the heatmap that indicates the regions of interest (regions in which the parasite was found).

Superposition: The heatmap obtained above is overlaid on the original image of the cell, thereby providing the doctors a visualization "rationale," which can be confirmed by their symptoms.

DATA DESCRIPTION AND PREPROCESSING

The dataset used comprises 27,558 red blood cell images obtained from the NIH database, consisting of equal numbers of parasitized and unparasitized cells amounting to 13,779 each. In preparation for utilizing the dataset for deep learning purposes, an extensive preprocessing approach is employed to ensure that the data is appropriately ready for training the neural network. The first step involves loading the data and converting it into RGB format, followed by resizing all images to have a fixed dimension of 64x64 pixels. Additionally, normalization of pixel values is carried out to limit the values within the range of [0,1] to accelerate the model's convergence rate. Moreover, to increase the robustness of the model against possible variances in imaging characteristics such as rotation and color depth, data augmentation is performed using transformations that randomly rotate images by up to 20 degrees, zoom images by 15%, and flip images horizontally and vertically.

EXPERIMENTAL DETAILS

The experiment was created within the framework of the Python language using the libraries of TensorFlow and Keras for building the CNN. All calculations were performed in Google Colab using a T4 GPU for speed up. The optimization algorithm employed for training our CNN was the Adam optimizer and the Binary Cross-entropy loss function. We used a batch size of 32, a dropout value of 0.5 for reducing overfitting, and an input image resolution of 64 x 64 pixels. To test the usefulness of the model in its intended application, a rigorous analysis of performance was conducted with measures like Precision, Recall, and F1-score, along with Confusion Matrix to detect any classification errors. After conducting statistical analysis, the application of Grad-CAM was done to understand how gradients from the last convoluted layer behaved. These gradients produced heat maps that were superimposed on top of the original micrograph to qualitatively check the focus of the model. In addition to the performance of the model, the experiment took into account the challenges associated with the use of this method in a clinical environment, such as working with noisy images due to variability in poor-quality microscopy. With the inclusion of Batch Normalization and Dropout layers, the architecture has been designed to maximize sensitivity with minimal false positives. In the last part of the experiment, it was shown that visual explanations provided by Grad-CAM were correlated with the actual location of the parasite Plasmodium.

RESULTS

The accuracy of the classifier exceeded 95%, where training and validation accuracies converged to 95.56% and 95.28% in the final epoch. The significant decrease in the validation loss of the model, which stands at 0.1299, indicates good convergence as well as a generalized model. This was confirmed using a confusion matrix and classification report, showing high precision and recall, hence ensuring that the system would be able to minimize false negatives, as is imperative when it comes to effective malaria screening. Moreover, the implementation of Grad-CAM resulted in qualitative success where generated heat maps were able to pinpoint the localization of the *Plasmodium* parasites in the red blood cells. It could be seen that the decisions made by the CNN were justified by the visual explanation since the heat maps depicted the dark, irregular spots typical for infections. Consequently, implementing Grad-CAM turned the model from a "black box" into a more transparent one that could easily be used in clinical practice as a tool for aiding the decision-making process.

SENSITIVITY ANALYSIS

In the first place, it is necessary to note the high sensitivity provided by the analysis, which is crucial for malaria testing. As a result, due to the high level of recall, the number of false negative classifications can be minimized, since the algorithm does not classify the infected samples among non-infected, which is an important factor in terms of the clinical use of the tool. Another confirmation of sensitivity is the integration of the Grad-CAM technique which allows verifying that the attention of the algorithm corresponds to the parasite inclusions in red blood cells, and not to some other details. Additionally, sensitivity was evaluated in case of different image variations, including the use of rotations, shifts, and zooms. Despite these factors, the architecture still managed to distinguish between parasitic and normal cells, which speaks about the generalization of the features of the *Plasmodium* infection that has been learned by the algorithm.

DISCUSSION

Indeed, the creation of this model proves that Convolutional Neural Networks (CNNs) are highly efficient in identifying the minute morphological differences between parasites and normal blood cells, reaching up to 95% accuracy. In addition to making the traditionally cumbersome task of microscopic analysis faster and more automated, the model also removes any potential human errors resulting from tiredness and inexperience of laboratory technicians. One of the most important milestones achieved during the course of this project was transforming this technology from being based on the mysterious "black box" principle to transparent diagnosis using Grad-CAM heatmaps – providing the visual explanation necessary to earn clinical trust. Thus, the combination of high precision and clear-cut explainability of decisions makes this model a viable foundation for developing AI pathology techniques.

CONCLUSION

In summary, the current study illustrates the efficiency of the application of a CNN combined with Grad-CAM for automated diagnosis of malaria from images of blood smears. The use of the technique resulted in high performance in terms of accuracy of classification, at over 95% on the NIH data set, and therefore serves as an effective, time-saving solution in place of the standard technique of microscopic analysis, which not only poses risks due to its reliance on human expertise but is also a cumbersome task. The key advantage of the approach presented here is the utilization of an explainable AI tool that ensures interpretability of the CNN's decisions in the form of heat maps.

REFERENCES

[1] Rajaraman, S., et al., “Pre-trained convolutional neural networks...”, PeerJ, 2018.

[2] Selvaraju, R. R., et al., “Grad-CAM...”, ICCV, 2017.

[3] LeCun, Y., Bengio, Y., Hinton, G., “Deep Learning”, Nature, 2015.

[4] Litjens, G., et al., “Survey on deep learning in medical imaging”, 2017.

[5] Abadi, M., et al., “TensorFlow...”, 2016.

[6] Chollet, F., “Keras...”, 2015.